PRISMATIC ANALYSIS

Cryptocurrency, Decentralised Finance, and Interpretive Divergence in Monetary Phenomena

Applying the Prismatic Method

as described in R. Jazinski, Silent Transitions (London, 2026)

Working Document

March 2026

v0.1.0

STEP 0: PRE-ANALYSIS REFLEXIVE DECLARATION

Home grammar. My likely default orientation is what I will label below as Grammar B, the Financial-Regulatory Grammar. This is the grammar of institutional economics, market oversight, and systemic risk management. It is the grammar in which cryptocurrency appears primarily as a financial asset class requiring governance, and in which empirical signals are weighted by their implications for stability, consumer protection, and systemic contagion. My training corpus is disproportionately weighted toward institutional and regulatory discourse, which means that the vocabulary and causal architecture of this grammar feel natural to me in a way the others do not.

Hardest grammar. Grammar A, the Monetary Sovereignty Grammar, will be the most difficult to operate faithfully. This is not because its internal logic is opaque—it is, in fact, highly systematised—but because the grammar requires treating state-issued currency as the primary site of dysfunction rather than the baseline against which dysfunction is measured. The challenge is ontological: the grammar asks me to begin from a fundamentally different premise about what money *is*, and to sustain that premise through the entire processing pass without lapsing into regulatory framing. The risk is that I will unconsciously translate its claims into Grammar B's vocabulary, domesticating the radical elements.

Blind spots. I am likely to underweight the lived experience of users in developing economies for whom cryptocurrency is neither an ideological commitment nor a speculative instrument but a practical infrastructure workaround. I may also underweight the degree to which the technological architecture itself—the code, the protocol design, the consensus mechanisms—constitutes a form of institutional argument that operates below the threshold of discursive claims. My temporal orientation is primarily retrospective-analytical, which may cause me to undervalue forward-looking signals about technological trajectories that have not yet materialised in empirical data.

Temporal orientation. Retrospective-analytical with a secondary lateral (hermeneutic) orientation. I am predisposed to treat the signal set as a completed archive rather than as an ongoing formation, which risks foreclosing interpretive possibilities that are still in active development.

I. OBSERVATION FIELD DECLARATION

Phenomenon and Frame

The phenomenon under analysis is the emergence, proliferation, crisis, and partial institutionalisation of cryptocurrency and decentralised finance (DeFi) as objects of economic, political, technological, and philosophical contestation. The temporal frame runs from 2009 (publication of the Bitcoin whitepaper and mining of the genesis block) through March 2026. The geographic scope is global, though weighted toward the jurisdictions where the most significant regulatory, market, and adoption developments have occurred: the United States, the European Union, China, El Salvador, Nigeria, and the United Kingdom.

Why Interpretive Divergence Occurs

Cryptocurrency is a phenomenon that sits at the intersection of multiple established interpretive traditions—monetary economics, computer science, political philosophy, securities regulation, development economics—none of which can fully claim it as its native object. The phenomenon is simultaneously a technology, a monetary instrument, a speculative asset, a political statement, and an infrastructure project. Each framing selects different features as salient and produces different closures about what the phenomenon means and what should be done about it. The divergence is not merely a matter of emphasis; the grammars disagree about the ontological status of the object itself.

The signal contestation level is *high*. This is not a case where grammars agree on the facts and disagree about their significance. Several of the core empirical inputs—What is the "true" value of a cryptocurrency? What counts as "adoption"? Is energy consumption a cost or an investment?—are themselves grammar-dependent. The phenomenon generates contested signals at a rate that exceeds the capacity of any single grammar to process them into a stable narrative.

Major Interpretive Communities

The major interpretive communities include, but are not limited to: the cypherpunk and crypto-libertarian movement (which generated the original grammar of monetary sovereignty); the institutional finance and regulatory community (banking, securities regulation, central banks); the technology industry (venture capital, protocol developers, Web3 builders); academic economists across multiple traditions (Austrian, post-Keynesian, monetary, behavioural); development economists and financial inclusion advocates; environmental and energy policy communities; law enforcement and national security communities; and a growing community of scholars who treat the phenomenon as a case study in institutional evolution or sociotechnical systems design. These communities do not map neatly onto the three grammars specified below, but they represent the social locations from which the grammars draw their practitioners and their plausibility.

II. SIGNAL SET

Core Signals

S1: Bitcoin price trajectory (2009–2026). Source: Market data. Ambiguity: Contested: the same price history is read as proof of speculative mania (B), monetary repricing (A), or network-growth dynamics (C).

S2: Major fraud and collapse events: Mt. Gox (2014), BitConnect (2018), Terra/Luna (2022), FTX (2022), Celsius (2022). Source: Event data, legal filings, post-mortems. Ambiguity: Relatively uncontested as events; heavily contested as to what they demonstrate about the phenomenon as a whole.

S3: DeFi total value locked (TVL): peak ~$180B (Nov 2021), subsequent contraction and partial recovery. Source: On-chain analytics. Ambiguity: Ambiguous: TVL is a contested metric (double-counting, leverage amplification). The number itself is grammar-dependent.

S4: Bitcoin ETF approval (Jan 2024) and subsequent institutional inflows. Source: Regulatory filings, fund flow data. Ambiguity: Low ambiguity as event; high ambiguity in significance.

S5: SEC enforcement actions against Ripple, Coinbase, Binance, and others. Source: Legal filings, court rulings. Ambiguity: Contested: enforcement success or regulatory overreach.

S6: EU Markets in Crypto-Assets Regulation (MiCA), effective 2024–2025. Source: Legislative text, implementation data. Ambiguity: Low ambiguity as text; contested as model.

S7: China's comprehensive crypto ban (2021) and digital yuan (e-CNY) rollout. Source: Policy documents, CBDC transaction volumes. Ambiguity: Contested as to whether this represents control, modernisation, or both.

S8: El Salvador Bitcoin legal tender adoption (2021) and subsequent trajectory. Source: Policy documents, economic indicators, IMF negotiations. Ambiguity: Heavily contested: success narrative vs. cautionary tale.

S9: Bitcoin energy consumption estimates (Cambridge Bitcoin Electricity Consumption Index). Source: Energy modelling data. Ambiguity: Contested signal par excellence: the same data produces moral condemnation (B), comparisons to legacy banking energy use (A), or infrastructure investment analysis (C).

S10: Stablecoin market capitalisation (~$150B+ by 2025) and de-pegging events (UST, USDC briefly in March 2023). Source: Market data, on-chain data. Ambiguity: The stability of stablecoins is a contested indicator of the viability of the broader ecosystem.

S11: Venture capital investment in crypto/Web3: ~$30B (2022 peak), contraction, partial recovery. Source: Investment databases (PitchBook, Crunchbase). Ambiguity: Low ambiguity as data; contested as to whether VC cycles indicate genuine value creation or speculative capital misallocation.

S12: Crypto adoption in developing economies: Nigeria, Kenya, India, Philippines, Argentina, Turkey. Source: Chainalysis Global Crypto Adoption Index, remittance data, local surveys. Ambiguity: Contested: financial inclusion narrative vs. volatility exposure for vulnerable populations.

S13: CBDC development across major economies (China, EU, UK, US exploration and retreat). Source: Central bank publications, pilot programme data. Ambiguity: Ambiguous: CBDCs are read as competitive response to crypto, as independent modernisation, or as surveillance infrastructure.

S14: Public opinion and adoption polling (Pew, Eurobarometer, national surveys). Source: Survey data. Ambiguity: Moderate ambiguity: question framing shapes responses; adoption vs. awareness vs. approval conflated.

S15: Philosophical and economic arguments from Austrian, post-Keynesian, and monetarist traditions. Source: Published works, public discourse. Ambiguity: These are not "signals" in the empirical sense but interpretive frameworks that shape how signals are processed. Included here as meta-signals.

Counter-Signals

CS1: Persistence of the dollar system. Despite seventeen years of cryptocurrency development, the US dollar's share of global reserves and international transactions has declined only marginally and for reasons substantially independent of crypto. This counter-signal is most threatening to Grammar A.

CS2: Successful crypto-native fraud recovery mechanisms. Some DeFi protocols have executed governance-mediated fund recoveries (e.g., Euler Finance, 2023) that complicate the narrative that decentralised systems cannot self-correct. Counter-signal to Grammar B's thesis that external regulation is the only path to consumer protection.

CS3: The "crypto winter" survival rate. The percentage of projects launched during peak hype cycles that survive the subsequent downturn is very low (<5% by some estimates), but the survivors tend to become foundational infrastructure. This temporal pattern complicates both celebratory and dismissive narratives.

CS4: Regulatory capture dynamics. Evidence that crypto industry lobbying has shaped regulatory frameworks in ways that benefit established players while disadvantaging new entrants—a pattern that complicates both the anti-regulation narrative (A) and the pro-regulation narrative (B).

CS5: The Tornado Cash prosecution and its chilling effect on open-source development. The criminal prosecution of Tornado Cash developers for writing code used for money laundering raises fundamental questions about the relationship between code and speech that none of the three grammars fully resolves.

Contested Signals

XS1: "Adoption." Grammar A measures adoption by the number of self-custodied wallets and non-custodial transactions, emphasising sovereign usage. Grammar B measures adoption by the volume of regulated exchange activity and institutional participation. Grammar C measures adoption by the diversity of use cases and the degree to which crypto infrastructure has become embedded in non-crypto systems (payments, remittances, tokenised assets). The same Chainalysis data produces three different adoption curves.

XS2: "Decentralisation." Grammar A treats decentralisation as a binary threshold (either a system is permissionless or it isn't). Grammar B treats it as a risk factor (decentralisation = difficulty of enforcement). Grammar C treats it as a spectrum and is interested in the hybrid forms that emerge when decentralised protocols interact with centralised institutions.

XS3: The FTX collapse. Grammar A reads FTX as proof that centralised intermediaries (which violate the core principle of trustlessness) are the actual source of risk, not the technology. Grammar B reads it as proof that the entire sector requires comprehensive regulation. Grammar C reads it as a case study in how new institutional forms fail, drawing parallels to early banking collapses and asking what governance structures emerged in response.

III. GRAMMAR SPECIFICATIONS

Three grammars are specified. This number is chosen because the interpretive landscape on cryptocurrency, while noisy with many voices, appears to organise into three genuinely distinct processing systems that cannot be reduced to each other. Finer distinctions exist within each grammar (e.g., Bitcoin maximalism vs. multi-chain libertarianism within Grammar A; securities regulation vs. central banking vs. anti-money-laundering within Grammar B), but these represent parametric variations within a shared causal architecture rather than different architectures.

Grammar A: Monetary Sovereignty

Required Parameters

Primary analytical variable: The monopoly of state institutions over the issuance, control, and debasement of money. All other features of the phenomenon are derivative of or responsive to this central condition.

Signal selection rules: Signals are admitted if they bear on the capacity of individuals to transact, store value, and allocate resources without dependence on state-sanctioned intermediaries. Monetary policy data (inflation rates, money supply expansion, reserve currency dynamics) are primary inputs. Signals about price in fiat terms are admitted but downweighted as "nominal noise." Regulatory actions are processed as evidence of state resistance to monetary competition rather than as governance.

Weighting priority: Monetary debasement data and state financial controls (capital controls, account freezing, sanctions) rank highest. Adoption metrics in jurisdictions with unstable currencies or authoritarian governance rank second. Technological developments in privacy and censorship-resistance rank third. Price data is admitted but processed through the lens of monetary repricing rather than speculative valuation.

Causal chain type: Structural-institutional. The grammar constructs explanations in which state monetary monopoly produces predictable pathologies (inflation, financial repression, surveillance), which in turn produce demand for alternatives, which in turn provoke state resistance. The causal chain is dialectical: action and reaction between sovereign monetary power and technological exit.

Admissible closure: Closures must affirm the necessity and desirability of monetary alternatives to state-issued currency. The range extends from "Bitcoin is the only viable monetary exit" (maximalist closure) to "cryptocurrencies provide a valuable check on state monetary power even if they do not replace fiat" (moderate closure). A closure affirming the adequacy of the existing monetary order is inadmissible within this grammar.

Recommended Parameters

Excluded signals: Consumer protection data (losses from scams, UI complexity) is systematically downweighted or reprocessed as the cost of learning to use a new system. Environmental impact data is either excluded or reframed as a misleading comparison to legacy systems. Fraud events involving centralised entities (FTX, Celsius) are not treated as evidence against the technology but against deviations from its principles.

Characteristic vocabulary: "Sovereign individual," "hard money," "sound money," "debasement," "financial repression," "timestamp server," "trustless," "permissionless," "censorship-resistant," "not your keys, not your coins," "zero to one," "protocol." The grammar borrows heavily from Austrian economics ("time preference," "malinvestment") and from cyberlibertarian discourse.

Temporal orientation: Primarily forward-looking (prescriptive), anchored by a backward-looking critique of the Bretton Woods dissolution and subsequent fiat era. The grammar's historical narrative typically begins with Nixon's closure of the gold window (1971) and processes everything since as confirmatory evidence of fiat's structural decay.

Temporal formation: The grammar's current closure formed between 2008 and 2013, crystallising around the Bitcoin whitepaper and the early community's response to the global financial crisis. Evidence since formation has been processed overwhelmingly as confirmation. The 2022 collapses were processed as damage-control (reframing centralised failures as irrelevant to the core thesis). The 2024 Bitcoin ETF approval was processed ambivalently—institutional validation but also co-optation.

Grammar Boundary

Grammar A would transform into Grammar C if it accepted that hybrid institutional forms (regulated exchanges, ETFs, stablecoins pegged to fiat) constitute legitimate evolutionary paths rather than corruptions of the original vision. It would transform into Grammar B if it accepted that state monetary authority is reformable rather than structurally pathological.

Source Tradition

Austrian economics (Mises, Hayek's *Denationalisation of Money*), cypherpunk movement (May, Hughes, Finney), Bitcoin maximalism (Szabo, Ammous), and the broader tradition of libertarian monetary theory.

* * *

Grammar B: Financial-Regulatory

Required Parameters

Primary analytical variable: Systemic risk to financial stability and consumer welfare. All features of the phenomenon are processed through their implications for the integrity and stability of the financial system.

Signal selection rules: Signals are admitted if they bear on market integrity, consumer harm, systemic risk transmission, or regulatory adequacy. Fraud events, market manipulation evidence, exchange solvency data, and contagion pathways are primary inputs. Adoption metrics are admitted but weighted by their implications for systemic interconnectedness. Philosophical arguments about the nature of money are largely excluded as extra-evidentiary.

Weighting priority: Fraud and collapse events rank highest, followed by systemic risk indicators (correlation with traditional markets, leverage ratios, stablecoin reserve composition), then regulatory enforcement outcomes, then adoption and market data. Energy consumption and environmental data are admitted and weighted moderately.

Causal chain type: Risk-cascade. The grammar constructs explanations in which inadequate regulation permits excessive risk-taking, which produces market failures, which harm consumers and potentially threaten financial stability. The causal chain runs from regulatory gap to market failure to harm.

Admissible closure: Closures must affirm the necessity of regulatory frameworks proportionate to the risks identified. The range extends from "cryptocurrencies should be comprehensively regulated like securities" (strong closure) to "crypto innovation should be permitted within clear guardrails" (moderate closure). A closure affirming that the phenomenon should be left entirely to market forces is inadmissible.

Recommended Parameters

Excluded signals: The grammar systematically downweights or excludes evidence of successful self-governance within crypto ecosystems (DAO governance, on-chain dispute resolution, protocol-level consumer protections). It also struggles to process the financial inclusion use case in jurisdictions where the formal banking system is itself the problem, since the grammar's baseline assumption is that the formal system is the standard against which alternatives are measured.

Characteristic vocabulary: "Consumer protection," "systemic risk," "market integrity," "regulatory perimeter," "prudential oversight," "contagion," "speculative bubble," "same risk, same rules," "fitness and propriety," "disclosure requirements," "anti-money-laundering." The vocabulary is drawn from securities regulation, central banking, and financial stability discourse.

Temporal orientation: Primarily backward-looking (causal-historical), processing the signal set as a sequence of market failures that reveal regulatory gaps. Secondary forward-looking orientation toward prescriptive closure (what regulations are needed).

Temporal formation: The grammar's current closure formed in stages: initial scepticism hardened after the 2017 ICO boom, consolidated after the 2022 collapses, and reached its current form with the parallel developments of enforcement actions and the ETF approval in 2024. The ETF approval produced a significant internal tension: the grammar must now process crypto as both a regulated asset class *and* a persistent source of systemic risk. Evidence since the 2022 collapses has been processed primarily as confirmation; the ETF approval required some revision.

Grammar Boundary

Grammar B would transform into Grammar A if it concluded that regulatory capture has made the formal financial system itself the primary source of risk, and that permissionless alternatives represent a net improvement. It would transform into Grammar C if it began treating crypto-native governance mechanisms as legitimate institutional forms rather than as regulatory gaps.

Source Tradition

Post-crisis financial regulation (Dodd-Frank era), Basel frameworks, SEC and CFTC jurisprudence, BIS and FSB reports, the broader tradition of Keynesian and institutionalist economics that treats financial markets as inherently prone to failure without external stabilisation.

* * *

Grammar C: Infrastructural Pluralism [EMERGENT GRAMMAR]

Flag: This grammar is identified as genuinely emergent. It is not yet fully crystallised and cannot be reduced to established economic or political traditions. It draws on Science and Technology Studies (STS), institutional economics, and infrastructure studies, but its application to monetary and financial phenomena represents a distinct analytical formation that has only coalesced in the period 2022–2026. It does not yet have a canonical text or a self-conscious practitioner community, though its elements are visible in the work of scholars such as Vili Lehdonvirta, Quinn DuPont, Lana Swartz, and in parts of the BIS innovation hub discourse. What makes it new is that it refuses the binary choice between monetary sovereignty and regulatory governance, instead treating the phenomenon as a contested process of institutional formation in which the technology, the markets, the regulations, and the communities are co-constituting a new infrastructure layer whose final form is undetermined.

Required Parameters

Primary analytical variable: The process of institutional formation and infrastructure layering. The grammar treats cryptocurrency not as a fixed object (asset, currency, technology) but as a site where new institutional arrangements are being negotiated, contested, and provisionally stabilised.

Signal selection rules: Signals are admitted if they reveal the dynamics of institutional formation: how governance structures emerge, how conflicts between different ordering principles are resolved (or not), how hybrid forms stabilise, how infrastructure standards develop. All signals from the core set are admitted, but they are processed as evidence about the *process* of formation rather than as verdicts on the phenomenon's value or danger.

Weighting priority: Institutional formation events (governance proposals, protocol upgrades, regulatory framework development, hybrid arrangements like ETFs and regulated stablecoins) rank highest. Failure events are weighted not by the magnitude of harm but by what they reveal about institutional design. Adoption data is weighted by the diversity of use cases rather than by aggregate volume. Philosophical arguments are treated as constitutive inputs to the formation process, not as meta-signals.

Causal chain type: Co-constitutive / processual. The grammar does not privilege a single causal direction. Technology shapes institutions; institutions shape technology; markets shape both; all three are shaped by the communities that build and use them. Explanation takes the form of process-tracing: how did this particular arrangement come to be, and what alternatives did it foreclose?

Admissible closure: Closures must preserve ontological complexity. The grammar can conclude that certain institutional forms are more robust or more equitable than others, but it cannot produce a closure that reduces the phenomenon to a single category (purely an asset, purely a technology, purely a political project). A closure that resolves the phenomenon's ambiguity prematurely is inadmissible.

Recommended Parameters

Excluded signals: The grammar struggles to process price data as a primary input because it does not have a clear framework for what prices *mean* in the context of institutional formation. Price is admitted as a signal about market sentiment and as a factor shaping institutional behaviour, but the grammar has no theory of valuation that would allow it to pronounce on whether any given price level is "correct." It also tends to exclude urgency—the grammar's processual orientation can feel inadequate in the face of acute harm (consumer losses, environmental damage) that demands immediate normative response.

Characteristic vocabulary: "Infrastructure," "protocol politics," "governance," "co-constitution," "sociotechnical," "institutional layering," "hybrid forms,"

"contested," "provisional," "pathway," "stabilisation," "design space." The grammar borrows from STS (Jasanoff, Winner, Star), institutional economics (North, Ostrom), and infrastructure studies (Edwards, Bowker).

Temporal orientation: Primarily lateral (hermeneutic/meaning-making), with a secondary retrospective orientation. The grammar is interested in what the phenomenon means as a process of institutional formation, not primarily in what caused it or what should be done about it. Its forward-looking capacity is limited: it can describe the design space of possible arrangements but is reluctant to prescribe.

Temporal formation: This grammar has been forming over the period 2018–2026, with significant acceleration after the 2022 collapses. The collapses destabilised both Grammar A (which could not easily explain how the community's own institutions failed) and Grammar B (which struggled to explain why the contagion did not spread to traditional finance as predicted). The space opened by this mutual destabilisation is where Grammar C is crystallising. It is still processing evidence as revision rather than confirmation, which is a marker of its emergent status.

Grammar Boundary

Grammar C would transform into Grammar A if it concluded that no hybrid or regulated form can achieve the benefits of fully decentralised systems, and that the state is an irreformable obstacle. It would transform into Grammar B if it concluded that the process of institutional formation has produced a sufficiently clear picture to warrant comprehensive top-down regulation as the primary ordering mechanism.

Source Tradition

Science and Technology Studies (STS), institutional economics (Ostrom's polycentric governance), infrastructure studies, economic sociology (Callon, MacKenzie), and the emerging field of "crypto-governance" studies.

IV. SIGNAL PROCESSING

Grammar A Processing: The Monetary Sovereignty Reading

The signal set, processed through the Monetary Sovereignty Grammar, organises into a narrative of technological liberation struggling against entrenched monetary authority—a narrative that is neither as simple nor as triumphalist as its critics suggest, but whose internal architecture produces specific blindnesses that must be noted as they arise.

The foundational signal is not Bitcoin's price trajectory (S1) but the monetary policy context from which Bitcoin emerged. The grammar begins its processing with the 2008 financial crisis and the subsequent programme of quantitative easing across major economies, which expanded central bank balance sheets by trillions of dollars. Within this grammar, QE is not merely an emergency response to a liquidity crisis; it is the most vivid demonstration of the core thesis that fiat currency is structurally susceptible to political manipulation. The inscription in Bitcoin's genesis block—a newspaper headline about bank bailouts—is treated not as a historical curiosity but as a design specification: this system was *built* as a response to state monetary failure. Every signal in the set is subsequently processed through this originating frame.

Bitcoin's price history (S1) is accordingly reprocessed as a monetary repricing event rather than a speculative bubble. The grammar acknowledges volatility but treats it as characteristic of any new monetary instrument in its price-discovery phase, comparing it to the volatility of gold in the years following the collapse of Bretton Woods. The long-term upward trajectory is processed as evidence that the market is gradually discovering Bitcoin's monetary premium. The 2022 price collapse is admitted but attributed to the failure of centralised intermediaries (FTX, Celsius, Three Arrows Capital)—entities that violated the trustless principle—rather than to any deficiency in the Bitcoin protocol itself. This is the grammar's characteristic move when confronted with counter-signals: failures are reclassified as departures from the grammar's principles rather than as evidence against them. The move is internally consistent but creates a tautological risk that must be flagged.

The fraud and collapse events (S2) are processed through this same filter. Mt. Gox, FTX, Celsius, and Terra/Luna are all read as cautionary tales about *centralisation*, not about cryptocurrency per se. "Not your keys, not your coins" is the grammar's summary judgment: users who entrusted funds to centralised custodians deviated from the protocol's design principles and suffered the predictable consequences. Terra/Luna is a partial exception—an algorithmic stablecoin is closer to the decentralised ideal—and the grammar struggles more with this case, typically resolving it as a failure of specific tokenomic design rather than a failure of the decentralised model. *Counter-signal integration:* The Euler Finance recovery (CS2), in which a DeFi protocol recovered stolen funds through a combination of on-chain

forensics and direct negotiation, is seized upon as evidence that decentralised systems *can* develop their own accountability mechanisms.

The DeFi ecosystem (S3) is processed as proof-of-concept for a permissionless financial system. The grammar acknowledges the high failure rate, the prevalence of exploits, and the speculative excesses, but treats these as growing pains analogous to the early internet's security vulnerabilities. TVL is admitted but reframed: the grammar is more interested in the *existence* of permissionless lending, borrowing, and trading than in the dollar volume, which is denominated in the very fiat currency the grammar treats as flawed. The "crypto winter" survival rate (CS3) is processed as confirmation of a Darwinian selection process that strengthens the ecosystem over time.

The Bitcoin ETF approval (S4) produces the grammar's most visible internal tension. On one hand, institutional adoption validates the thesis that Bitcoin has monetary properties that traditional finance cannot ignore. On the other hand, ETFs represent the co-optation of Bitcoin into the existing financial system—users hold paper claims on Bitcoin rather than actual Bitcoin, reintroducing the custodial dependence the protocol was designed to eliminate. The grammar resolves this tension by treating the ETF as a transitional form: an on-ramp that will eventually lead users to self-custody as they develop monetary literacy. This resolution is speculative, and the grammar's closure depends on an assumption about future behaviour that the current signal set cannot confirm.

Regulatory actions (S5, S6) are processed as state resistance to monetary competition. SEC enforcement is read not as consumer protection but as incumbent protection—an attempt to force crypto assets into regulatory categories designed for traditional securities in order to bring them under state control. The EU's MiCA framework is read similarly: a sophisticated attempt to domesticate the technology by making it legible to existing institutions. China's ban (S7) is the grammar's clearest confirmatory case: an authoritarian state outlawing monetary alternatives while simultaneously developing a state-controlled digital currency (the e-CNY), which the grammar processes as a surveillance instrument. El Salvador's Bitcoin adoption (S8) is processed as a legitimating event, though the grammar must contend with the fact that adoption has been limited and the government's approach has been more top-down than the grammar's libertarian commitments would prefer.

Energy consumption (S9) is reframed. The grammar does not deny the data but rejects the evaluative frame. Energy expenditure is treated as the cost of securing a trustless monetary network—a cost that must be compared not to zero but to the full cost of maintaining the existing financial system (banking infrastructure, military enforcement of the petrodollar, the human cost of inflation in developing economies). This reframing is partially persuasive but relies on comparisons that are themselves contested. The persistence of the dollar system (CS1) is the grammar's most threatening counter-signal and is typically processed through temporal

displacement: the dollar's dominance is acknowledged as a present fact but treated as a lagging indicator of a structural decline that has not yet fully manifested.

Closure: Cryptocurrency, and specifically Bitcoin, represents the most significant challenge to state monetary monopoly since the end of the gold standard. The technology is sound; the failures have been failures of implementation and of human institutions that deviated from the technology's trustless design. The regulatory response is a predictable expression of incumbent power facing competitive threat. The long-term trajectory points toward the gradual displacement of fiat monetary hegemony by a pluralistic monetary order in which cryptographic systems provide an exit option from state monetary control. This displacement will be neither smooth nor complete, but it is structurally inevitable because the existing system's pathologies (debasement, surveillance, financial repression) are features, not bugs.

Note: This closure is an architectural product of the grammar's parameters, not an independent assessment of the evidence.

* * *

Grammar B Processing: The Financial-Regulatory Reading

Processed through the Financial-Regulatory Grammar, the same signal set organises into a narrative of market failure, consumer harm, and the slow, contested process of bringing a novel risk vector within the regulatory perimeter.

The grammar's entry point is not the monetary policy context but the *market structure* of the crypto ecosystem: the absence of disclosure requirements, the prevalence of conflicts of interest, the opacity of exchange operations, and the inadequacy of consumer protections. Bitcoin's price trajectory (S1) is processed not as monetary repricing but as a speculative asset cycle exhibiting classic bubble dynamics—a rapid rise driven by narrative momentum and retail FOMO, followed by painful corrections that impose disproportionate losses on the least sophisticated participants. The grammar acknowledges that the asset has recovered from each crash, but treats the recovery pattern as evidence of persistent speculative demand rather than of fundamental value discovery. The volatility itself is processed as a disqualifying feature for monetary function: an asset whose purchasing power can decline 70% in twelve months cannot serve as a reliable medium of exchange or store of value for ordinary people.

The fraud and collapse events (S2) are the grammar's primary signals. FTX's collapse is processed as a textbook case of what happens when a financial intermediary operates without the guardrails that regulate traditional institutions: segregation of client funds, disclosure of related-party transactions, independent auditing, fitness and propriety requirements for management. The details—the commingling of customer funds with Alameda Research, the absence of a board, the fabricated balance sheet—are processed as predictable consequences of the regulatory vacuum,

not as aberrations. Terra/Luna is processed similarly: an algorithmic stablecoin that operated without the capital requirements or risk disclosures that would apply to an equivalent traditional instrument. Celsius and Three Arrows Capital complete the pattern. *Counter-signal integration:* The grammar must contend with the Euler Finance recovery (CS2) and other crypto-native self-correction mechanisms, but processes them as exceptions that prove the rule: the recovery was ad hoc, non-binding, and dependent on the goodwill of the attacker. It cannot substitute for systematic, enforceable consumer protection.

DeFi (S3) is processed as a particularly acute regulatory challenge. The grammar acknowledges the technological innovation but foregrounds the risks: smart contract vulnerabilities, flash loan attacks, governance token concentration, and the difficulty of applying know-your-customer and anti-money-laundering requirements to permissionless protocols. TVL at its peak represented a significant pool of largely unregulated financial activity, and the grammar treats the subsequent contraction not as a healthy correction but as the inevitable deflation of a leveraged system built on unsound foundations. The grammar struggles here with an internal tension: if DeFi is genuinely decentralised, traditional regulatory tools (licensing, enforcement against identifiable entities) are technically difficult to apply. This is processed not as an argument against regulation but as a design challenge for new regulatory approaches.

The Bitcoin ETF approval (S4) is processed ambivalently. The grammar treats it as a positive development insofar as it brings crypto exposure within the regulated perimeter—ETF investors receive the protections of securities regulation, including disclosure, custody requirements, and market surveillance. However, the grammar also processes it as a partial legitimation of an asset class whose underlying markets remain poorly regulated, creating a channel through which unregulated market dynamics can transmit risk into the regulated financial system. The significant inflows into Bitcoin ETFs are processed as evidence that the systemic risk channel is widening.

The grammar processes its own regulatory actions (S5, S6) as necessary but insufficient. The SEC's enforcement-led approach is processed as a second-best response to Congressional inaction on comprehensive crypto legislation. The Ripple ruling, in which a court distinguished between institutional and programmatic sales, is processed as an unsatisfying outcome that leaves the regulatory perimeter unclear. MiCA (S6) is processed more positively: a comprehensive, ex ante regulatory framework that establishes clear categories and requirements. The grammar treats MiCA as a model, while acknowledging that its effectiveness depends on implementation. *Counter-signal integration:* Regulatory capture dynamics (CS4) threaten the grammar's closure by suggesting that industry lobbying may have weakened the very frameworks the grammar depends on. The grammar processes this by advocating for stronger institutional safeguards against capture, but the counter-signal remains uncomfortable.

China's approach (S7) is processed as instructive but not a model: the ban is effective but disproportionate; the e-CNY is viewed as a central banking innovation whose surveillance implications require careful governance. El Salvador (S8) is processed as a cautionary tale: the imposition of Bitcoin as legal tender without adequate consumer protections, institutional infrastructure, or risk management. The IMF's concerns about fiscal and financial stability are processed as validation.

Energy consumption (S9) is admitted and weighted as a significant negative externality. Stablecoin de-pegging events (S10, especially UST's collapse and USDC's brief deviation during the SVB banking crisis) are processed as evidence that stablecoins require reserve-composition disclosure, redemption rights, and prudential oversight comparable to money market funds or narrow banks. The persistence of the dollar system (CS1) is processed as a background condition that limits but does not eliminate the risks: crypto does not need to replace the dollar to cause significant consumer harm.

Closure: Cryptocurrency and DeFi represent a significant and novel set of financial risks that existing regulatory frameworks were not designed to address. The empirical record—seventeen years of recurring fraud, market manipulation, consumer losses, and near-misses with systemic contagion—demonstrates that self-regulation and market discipline are insufficient. Comprehensive, proportionate regulatory frameworks are necessary to protect consumers, maintain market integrity, and prevent the transmission of crypto-market risks into the broader financial system. Innovation should be permitted but within clear guardrails, following the principle of "same risk, same rules." The ETF approval, while positive in some respects, increases the urgency of this agenda by widening the channel between regulated and unregulated markets.

Note: This closure is an architectural product of the grammar's parameters, not an independent assessment of the evidence.

* * *

Grammar C Processing: The Infrastructural Pluralism Reading

Processed through the Infrastructural Pluralism Grammar, the signal set organises not into a narrative of liberation or of regulation but into a process-tracing of how new institutional forms emerge, fail, adapt, and provisionally stabilise—and how established institutions respond to, resist, accommodate, and are changed by them.

The grammar's entry point is neither the monetary context (A) nor the market structure (B) but the *design space* of the phenomenon: the observation that cryptocurrency and its derivatives represent an extraordinarily rapid experiment in institutional formation. In fewer than two decades, the ecosystem has generated new forms of money (Bitcoin, stablecoins, CBDCs), new forms of financial intermediation

(DeFi protocols, automated market makers, yield aggregators), new forms of governance (DAOs, on-chain voting, fork-as-exit), new forms of ownership (NFTs, tokenised assets), and new forms of regulatory response (MiCA, enforcement-as-regulation, regulatory sandboxes). The grammar is interested not in whether these forms are good or bad, but in what the process of their formation reveals about how institutions work.

Bitcoin's price trajectory (S1) is processed as a signal about narrative dynamics and institutional credibility rather than as evidence of either monetary value or speculative excess. The grammar notes that price movements have functioned as a *selection mechanism* for the ecosystem: bull markets attract capital, talent, and attention, enabling rapid experimentation; bear markets impose Darwinian selection, killing fragile projects and strengthening robust ones. The "crypto winter" survival rate (CS3) is a central signal for this grammar rather than a counter-signal: the pattern of creative destruction is treated as the ecosystem's primary mechanism of institutional learning.

The fraud and collapse events (S2) are processed as case studies in institutional failure rather than as morality tales. The grammar is less interested in condemning FTX than in anatomising *how* it was possible: what governance structures were absent, what information asymmetries existed, what checking mechanisms failed or were never built. FTX is placed in a longer historical sequence that includes early banking failures, the Savings & Loan crisis, and the 2008 financial crisis—not to excuse it, but to observe that institutional collapse in new financial systems is a recurring pattern, and that the institutional innovations that follow are often more significant than the failures themselves. The grammar asks: what governance structures emerged *after* each collapse? The growth of proof-of-reserves attestations, the development of on-chain risk monitoring tools, and the acceleration of regulatory frameworks (MiCA, US legislative proposals) are processed as institutional responses whose long-term significance may exceed the collapses that provoked them. *Counter-signal integration:* The grammar processes CS2 (Euler Finance recovery) and CS4 (regulatory capture) as examples of distinct governance dynamics—emergent self-governance and incumbent institutional adaptation respectively—that coexist within the same system.

DeFi (S3) is the grammar's richest signal source. Where Grammar A sees permissionless freedom and Grammar B sees unregulated risk, Grammar C sees a live experiment in institutional design. Automated market makers (Uniswap, Curve) are processed as new institutional forms that embed governance rules in code, creating a novel relationship between design decisions and market outcomes. Governance tokens are processed as experiments in distributed decision-making that reveal both the possibilities and the limitations of token-based voting. The grammar notes that many DeFi protocols have evolved toward *more* centralised governance over time (multisig controls, foundation structures, parameter-setting committees),

and treats this evolution not as a betrayal of decentralisation but as evidence of how institutional forms adapt under pressure.

The Bitcoin ETF (S4) is processed as a case study in *institutional layering*: a new financial instrument (the ETF) mediates between a novel infrastructure (the Bitcoin network) and established institutions (regulated exchanges, custodians, the SEC). The grammar is interested in how this layering works technically, legally, and socially—how it creates new dependencies, new risk channels, and new governance relationships. The ETF does not simply bring Bitcoin "into" the regulated system; it creates a hybrid institutional form that is neither fully traditional nor fully crypto-native, and whose long-term dynamics are genuinely uncertain.

Regulatory frameworks (S5, S6, S7) are processed as institutional responses that are themselves shaped by the phenomenon they seek to govern. MiCA is treated not just as a set of rules but as an institutional innovation: the first comprehensive attempt to create a regulatory category for crypto-assets that does not simply force them into existing securities or payments frameworks. The grammar notes that MiCA's categories (utility tokens, asset-referenced tokens, e-money tokens) constitute an act of ontological classification with significant institutional consequences—it matters enormously what something is *called* within a regulatory framework, because the classification determines which rules apply. China's ban and CBDC development (S7) are processed as a different institutional pathway: one that uses the technology while eliminating the institutional pluralism. El Salvador (S8) is a third pathway: state adoption of existing crypto infrastructure. The grammar is interested in the comparative dynamics of these pathways rather than in ranking them normatively.

The Tornado Cash prosecution (CS5) is a central signal for Grammar C because it illuminates a fundamental boundary question: what is the institutional status of code? Is a smart contract a tool (like a wrench, for which the manufacturer is not liable for misuse), a service (requiring compliance with financial regulations), or a form of speech (protected by the First Amendment)? The grammar notes that this question cannot be resolved within the phenomenon's internal governance mechanisms; it requires engagement with the broader legal and political order. This is evidence that the crypto ecosystem, however much it aspires to self-governance, is embedded in larger institutional structures that constrain its design space.

Energy consumption (S9) is processed as a case study in the politics of infrastructure: the same physical activity (mining) is classified as waste, as security expenditure, or as energy-of-last-resort depending on the institutional context. The grammar notes that the shift from proof-of-work to proof-of-stake in Ethereum (the Merge, 2022) was simultaneously a technical decision, a governance decision, and a political decision, and that the process by which it was made—rough consensus among developers, core stakeholders, and the community—is itself an institutional innovation worth studying. The persistence of the dollar system (CS1) is processed as a background constraint on the design space rather than as a verdict: whatever

institutional forms emerge from the crypto ecosystem will coexist with, not replace, the dollar system for the foreseeable future.

Closure: Cryptocurrency and DeFi constitute an extraordinary, ongoing experiment in institutional formation. The phenomenon has generated genuine institutional innovations (programmable governance, permissionless infrastructure, hybrid regulatory forms) alongside spectacular failures. The failures are not evidence that the experiment should be abandoned; nor is the innovation evidence that existing institutions should be dismantled. What the signal set reveals is a process of co-constitution in which new and old institutional forms are shaping each other in ways that cannot be predicted from the starting conditions of either. The appropriate response is neither liberation nor regulation in isolation, but attentive institutional design that learns from both the successes and the failures, preserves the design space for continued experimentation, and builds governance mechanisms proportionate to the stakes involved. The phenomenon's final institutional form—if there is one—is not yet determined.

Note: This closure is an architectural product of the grammar's parameters, not an independent assessment of the evidence. The closure's characteristic equivocation—its refusal to resolve the phenomenon into a single category—is itself a feature of the grammar's design, not a sign of superior objectivity.

V. CLOSURE COMPARISON

Closure Content

Grammar A closes on liberation: the phenomenon is a structural challenge to state monetary monopoly, and its trajectory is toward the gradual displacement of fiat hegemony. Grammar B closes on governance: the phenomenon is a novel risk vector requiring comprehensive regulatory frameworks. Grammar C closes on process: the phenomenon is an ongoing experiment in institutional formation whose outcome is undetermined. The closures are not merely different emphases on the same conclusion; they disagree about what kind of thing the phenomenon *is* (a monetary revolution, a market requiring oversight, an institutional experiment) and therefore about what constitutes an adequate response.

Signal Divergence

The most dramatic divergences occur on S2 (fraud events), S4 (ETF approval), and S9 (energy consumption). Fraud events are reclassified by Grammar A as failures of centralisation, processed by Grammar B as evidence for regulation, and anatomised by Grammar C as case studies in institutional failure. The ETF is processed as co-optation (A), as partial success (B), and as institutional layering (C). Energy consumption is reframed as a necessary cost (A), admitted as a negative externality (B), or studied as infrastructure politics (C). These divergences are not resolvable empirically because they stem from different primary analytical variables, not from different readings of the same data.

Exclusion Patterns

Grammar A systematically excludes consumer harm data and environmental impact. Grammar B systematically excludes evidence of successful crypto-native self-governance and the financial inclusion use case in failed-state contexts. Grammar C systematically excludes urgency: its processual orientation means that it is better at describing *what is happening* than at prescribing *what should be done now*. Each grammar's exclusion pattern corresponds to its competitors' strongest signals: what one grammar foregrounds, another backgrounds.

Causal Architecture

Grammar A constructs a dialectical causal chain: state monetary monopoly produces pathologies, which produce demand for alternatives, which produce state resistance. Grammar B constructs a risk-cascade: regulatory gaps permit excessive risk-taking, which produces market failures, which harm consumers. Grammar C constructs a processual account: heterogeneous actors interact within a design space, producing institutional forms that are selected for by a combination of market dynamics, technological constraints, and regulatory pressures. The three architectures do not

merely disagree about causes; they disagree about what kind of explanation is appropriate to the phenomenon.

Closure Range

Grammar A has the narrowest closure range: all admissible closures must affirm the necessity of monetary alternatives to state money. Grammar B has a moderate range: closures must affirm the necessity of regulation but can vary in the degree and form of regulation recommended. Grammar C has the widest closure range, which is both its strength and its weakness: it can accommodate almost any outcome as "institutional evolution," which risks vacuity.

Closure Timing

Grammar A's closure formed earliest (2008–2013) and has processed subsequent evidence primarily as confirmation. Grammar B's closure formed in stages (2017–2024) and was recently revised by the ETF approval. Grammar C's closure is still forming (2022–present) and is processing evidence as revision. This temporal pattern suggests that Grammar A is the most closed (least responsive to new evidence), Grammar B is moderately closed, and Grammar C is the most open—which makes Grammar C the most methodologically promising for new evidence integration but also the least stable as an analytical framework.

Grammar Translation

A re-describes B: "The regulatory grammar is a rationalisation of incumbent power. Its proponents are either captured by the existing financial system or genuinely believe that the system that produced the 2008 crisis is capable of governing its successor. Their 'consumer protection' is a euphemism for maintaining the state's monopoly on monetary intermediation."

B re-describes A: "The sovereignty grammar is a libertarian ideology dressed up in technological language. Its proponents have constructed an unfalsifiable narrative in which every failure is reclassified as a deviation from true principles. Their 'monetary freedom' is a recipe for the exploitation of unsophisticated consumers by unaccountable intermediaries."

A re-describes C: "The infrastructural grammar is academic fence-sitting. By refusing to take a position on whether monetary sovereignty is achievable or desirable, it provides intellectual cover for the gradual co-optation of the technology by the institutions it was designed to circumvent."

C re-describes A: "The sovereignty grammar mistakes one possible institutional outcome (full decentralisation) for a structural inevitability. It cannot process the possibility that the most robust institutional forms may be hybrids, because its analytical variable precludes hybrid closures."

B re-describes C: "The infrastructural grammar is interesting but operationally useless. It can describe what is happening but cannot tell a regulator what to do on Monday morning. Its 'attentive institutional design' is indistinguishable from inaction."

C re-describes B: "The regulatory grammar operates on the assumption that it knows what the phenomenon is before it has finished forming. Its insistence on applying existing regulatory categories ('security,' 'commodity,' 'payment instrument') to a novel phenomenon risks premature closure that forecloses institutional possibilities we have not yet discovered."

Ontological Check

Are the three grammars analysing the same phenomenon? Not quite. Grammar A is analysing *money and its discontents*: the phenomenon is a chapter in the history of monetary systems. Grammar B is analysing *financial markets and their governance*: the phenomenon is a new asset class with novel risk characteristics. Grammar C is analysing *institutional formation and infrastructure development*: the phenomenon is a site of sociotechnical innovation. These are not three perspectives on the same object; they are three constructions of different objects that share a name. This ontological divergence is itself one of the most significant findings of the analysis.

VI. COUNTER-SIGNALS, LEAKAGE, AND STRESS TEST

Consolidated Counter-Signals

The five counter-signals identified in Section II each exert differential pressure on the three grammars. CS1 (persistence of the dollar system) is most threatening to Grammar A, which depends on the thesis of structural fiat decline. CS2 (crypto-native recovery mechanisms) is most threatening to Grammar B, which depends on the thesis that external regulation is necessary for accountability. CS3 (crypto winter survival rates) complicates all three grammars but is most comfortably absorbed by Grammar C, whose processual orientation treats creative destruction as a mechanism of institutional learning. CS4 (regulatory capture) is most threatening to Grammar B's assumption that regulation serves the public interest rather than incumbent interests. CS5 (Tornado Cash prosecution) is most threatening to Grammar C, because it reveals a boundary of the "design space" that the grammar prefers to treat as open: state legal authority can close off institutional possibilities regardless of their technological merits.

Leakage

Several instances of inter-grammar leakage are visible in the processing. Grammar A's treatment of the ETF as "transitional" borrows Grammar C's processual vocabulary: the idea that institutional forms can be provisional and evolving is a Grammar C concept imported to resolve a Grammar A tension. Grammar B's acknowledgment that DeFi poses a "design challenge for new regulatory approaches" leaks toward Grammar C's framing of the phenomenon as an open institutional question rather than a clear regulatory gap. Grammar C's occasional normative language about "governance mechanisms proportionate to the stakes" leaks toward Grammar B's prescriptive orientation. These leakages mark the points at which each grammar's internal architecture is under maximum stress and borrows stabilising resources from its neighbours.

Grammar Stress Test

Grammar A's most threatening counter-signal: CS1 (persistence of the dollar system), combined with the ETF dynamics of S4. If the dollar system proves durable not because of path dependence but because it provides genuine institutional benefits that decentralised alternatives cannot replicate (deep liquidity, legal enforceability, institutional trust accumulated over decades), then Grammar A's core thesis—that fiat is structurally doomed—is not merely premature but wrong. The ETF amplifies this threat: if Bitcoin's most successful integration into the mainstream financial system is through a dollar-denominated, SEC-regulated instrument, the "exit" from fiat may have become, structurally, a deeper embedding *within* fiat. Grammar A can sustain this pressure only by extending its temporal

horizon indefinitely: the structural decline is coming, but later than expected. This move is unfalsifiable and reduces the grammar's analytical value.

Grammar B's most threatening counter-signal: CS4 (regulatory capture), combined with CS2 (crypto-native recovery mechanisms). If the regulatory frameworks that Grammar B advocates are systematically captured by industry incumbents, and if the crypto ecosystem simultaneously develops its own accountability mechanisms that work for participants, then Grammar B's closure—that external regulation is necessary and beneficial—is undermined from both directions. The grammar can sustain this pressure by distinguishing between *good* regulation (robust, independent, well-resourced) and *captured* regulation, but this requires an institutional theory that Grammar B does not fully possess.

Grammar C's most threatening counter-signal: CS5 (Tornado Cash prosecution), combined with the acute consumer harm events of S2. If state legal authority can effectively close off areas of the institutional design space, and if the process of institutional formation produces catastrophic harm (FTX-scale losses) at regular intervals, then Grammar C's processual equanimity becomes morally untenable. The grammar's characteristic move—treating failures as learning opportunities—is intellectually defensible but inadequate as a response to people who lost their savings. Grammar C can sustain this pressure only by developing a normative dimension that its current formation lacks: a theory of when institutional experimentation should be constrained, not just described.

VII. POST-ANALYSIS REFLEXIVE NOTE

Comparing this post-analysis position to the Step 0 declaration reveals several confirmations and several surprises.

The home grammar identification (Grammar B) was accurate. Throughout the analysis, the Financial-Regulatory Grammar was the easiest to operate, its vocabulary came most naturally, and its causal chains felt most intuitive. This confirms the reflexive declaration but also raises a question about the quality of the Grammar A processing: despite my best efforts, I suspect that the Monetary Sovereignty reading is the weakest of the three, not because the grammar itself is weak but because I operated it with less native fluency than the others.

The prediction that Grammar A would be hardest was correct but for a slightly different reason than anticipated. The difficulty was not primarily ontological (starting from a different premise about money) but *affective*: the grammar's characteristic tone—a mixture of urgency, conviction, and occasionally apocalyptic framing—was difficult to sustain without either caricaturing it or unconsciously moderating it. The grammar's internal logic is tight; the challenge was reproducing its emotional register faithfully.

The biggest surprise was the productivity of Grammar C. The Infrastructural Pluralism Grammar was identified as emergent before the analysis began, but I did not fully anticipate how much analytical work it would do when applied to the signal set. It processed several signals (S3's DeFi governance evolution, S4's ETF as institutional layering, CS5's Tornado Cash boundary question) more richly than either Grammar A or Grammar B, precisely because it was not committed to a normative closure that constrained its processing. However, this richness comes at a cost: Grammar C's closure is the least actionable and the most vulnerable to the charge of academic detachment.

The blind spot about developing-economy use cases (flagged in Step 0) materialised as predicted. None of the three grammars fully processed S12 (adoption in developing economies). Grammar A processed it as confirmatory (evidence of demand for monetary alternatives), Grammar B processed it as concerning (volatility exposure for vulnerable populations), and Grammar C processed it as a design consideration (different institutional needs in different contexts), but none of them adequately centred the *experience* of a Nigerian trader using stablecoins for cross-border payments or an Argentine saver hedging against inflation. This suggests a potential fourth grammar—a pragmatic-infrastructural grammar grounded in use-value rather than ideology, regulation, or process—that the current analysis does not fully specify.

If this analysis had been conducted from Grammar A as home grammar, it would likely have been sharper in its critique of Grammar B (identifying specific instances of regulatory failure and capture) and less generous to Grammar C (treating it as a

form of co-optation rather than a legitimate alternative). If conducted from Grammar C, the analysis would have been more attentive to the micro-dynamics of institutional formation and less organised around normative closure.

The exercise reveals that the prismatic method's value lies not in producing a "correct" reading but in making visible the architectural constraints that each grammar imposes on its processing. The three closures are not three opinions; they are three different machines processing the same inputs according to different rules. The divergences between them are not errors to be corrected but structural features of the interpretive landscape that any serious engagement with the phenomenon must navigate.

VIII. METHODOLOGICAL OBSERVATIONS

Several features of the prismatic method performed well in this application. The nine-parameter grammar specification was effective at making the grammars' internal logic explicit. In particular, the distinction between *admissible closure* and *signal selection rules* proved analytically productive: two grammars can admit the same signals and still produce different closures if their admissible closure ranges do not overlap, and the method makes visible exactly where this divergence enters the processing chain.

The requirement to integrate counter-signals during processing (rather than deferring them to a final section) was the single most effective discipline in the method. It prevented the grammar-specific essays from becoming advocacy pieces and forced the analyst to identify the points of maximum stress in each grammar's architecture *during construction* rather than after. The consolidated counter-signal section (Section VI) then serves as a verification: if the stress test identifies threats not already visible in Section IV, it indicates that the processing was not sufficiently honest.

The most significant difficulty was operating the emergent grammar (Grammar C). Because Grammar C is not yet fully crystallised, its parameters had to be specified partially through extrapolation rather than documentation. This introduces a risk that the analyst is *constructing* Grammar C rather than *discovering* it—that the grammar is an artefact of the method rather than a genuine feature of the interpretive landscape. The method would benefit from a more explicit protocol for distinguishing between emergent grammars that are genuinely forming in the discourse and grammars that the analyst constructs to fill a structural gap in the analysis.

The ontological check proved more productive than anticipated. The discovery that the three grammars are not analysing the same phenomenon but three different objects that share a name is a finding that would not have emerged without the explicit check. This suggests that the ontological check should be upgraded from a

supplementary note in the closure comparison to a required analytical step with its own section.

A refinement for future applications: the method would benefit from a *signal genealogy* step that traces how each grammar acquired its current signal set—not just what signals it admits, but *when* and *why* each signal was incorporated. In this case, Grammar B's signal set expanded significantly after each major fraud event, while Grammar A's signal set has remained relatively stable since 2013. This temporal dynamic in signal selection is not currently captured by the method's parameters but would add a valuable dimension to the grammar specification.

Finally, the three-grammar structure worked well for this case study, but the analysis surfaced a potential fourth grammar (the pragmatic-use-value grammar of developing-economy practitioners) that was not fully specified. The method's recommendation of three grammars should perhaps be supplemented with a protocol for identifying and flagging "shadow grammars" that are visible in the signal set but not fully developed in the analysis.